

Two Reliability Philosophies under Disaster-Grade Networks:

A Two-Layer Benchmark of `rmw_zenoh` and CycloneDDS

for ROS 2 Search-and-Rescue Swarms

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Abstract—Middleware benchmarks for ROS 2 report performance almost exclusively under ideal network conditions, yet search-and-rescue (SAR) deployments operate on degraded links. We present a two-layer benchmark that measures `rmw_zenoh` and CycloneDDS under physically emulated degradation (`tc netem`): a transport echo microbenchmark and a complete simulated SAR mission with four UAVs. Across $2 \times 6 \times 5$ runs we find that the two middlewares fail *differently* rather than one dominating: under 30 % packet loss CycloneDDS converts loss into a uniform ~ 2.3 s latency wall and completes 5/5 missions, while `rmw_zenoh` delivers the majority of samples faster but sheds 35 % of them and completes 1/5; in the SAR-realistic combined condition (200 ms + 15 % loss) their RTT distributions cross, with Zenoh ahead on the median and behind on the tail. Transport tails degrade by up to three orders of magnitude while mission time varies by at most 12 %, showing that swarm autonomy compresses transport degradation up to an RMW-specific failure threshold—an effect invisible to transport-only benchmarks. We release the full benchmark, scenarios, and data for reproduction.

Index Terms—ROS 2, middleware, Zenoh, DDS, search and rescue, network degradation, multi-robot systems, benchmark

I. INTRODUCTION

Communication is the weakest link of field robotics: disaster environments combine long relays, interference, and non-line-of-sight propagation into links with hundreds of milliseconds of latency and double-digit packet loss [1]. ROS 2 delegates transport to interchangeable middleware implementations (RMW) [2], and the recent `rmw_zenoh` offers an alternative to the DDS family designed for constrained and wide-area networks [3]. Existing comparisons, however, measure throughput and latency on healthy networks [4]–[6], leaving open the question that matters operationally: *what does middleware choice cost a mission when the network degrades?*

We answer it with a two-layer benchmark. The transport layer measures round-trip time (RTT) and end-to-end sample loss over an echo channel; the mission layer runs a complete simulated SAR operation (four UAVs, ground control station, area coverage, victim detection) over the same degraded

interface. Degradation is applied *physically* with `tc netem` [7], so measured differences belong to the middleware alone.

Contributions: (i) the first two-layer ROS 2 middleware benchmark under controlled physical degradation, including the combined latency+loss condition absent from prior work; (ii) evidence that CycloneDDS and `rmw_zenoh` embody two distinct reliability philosophies with regime-dependent, qualitatively different failure modes; (iii) a quantified transport-to-mission compression effect (tails degrade $\sim 10^3\times$, mission time $\leq 12\%$) with RMW-specific failure thresholds; (iv) a reproducible open infrastructure (scenarios, manifests, analysis).

II. RELATED WORK

Early ROS 2 evaluations characterized DDS performance and executor overheads [5], [6], [8]. Liang et al. [4] compared Zenoh, MQTT, Kafka, and DDS on throughput and latency, finding Zenoh competitive—but exclusively on healthy networks. Disaster-robotics literature documents degraded field links and their operational cost [1], yet does not connect them to middleware selection. To our knowledge no prior work measures ROS 2 RMW implementations under controlled loss/latency at *mission* level; this gap is the target of the present study.

III. SYSTEM AND METHOD

A. Two-layer design

Fig. 1 shows the architecture. Both layers run unmodified over either RMW, selected via `RMW_IMPLEMENTATION`; the Zenoh router is started before the nodes. Degradation is applied on the shared interface so both layers experience identical network physics.

B. Transport microbenchmark

A client publishes timestamped payloads (64 B command class, 4 KB telemetry class, 64 KB map/image class) echoed by a server on the same host; RTT_i uses a single clock. Loss is the fraction of samples not returned within a per-sample deadline. We report p50/p95 per (RMW, condition) over $N=5$ repetitions of 20 s per payload class.

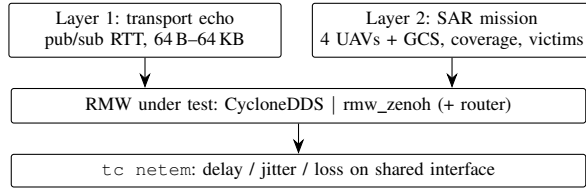


Fig. 1. Two-layer benchmark: identical nodes and traffic over either middle-ware; physical degradation under both layers.



Fig. 2. Simulated SAR scene: four UAVs covering a disaster area (Gazebo); the GCS dashboard and RViz operator view run over the degraded link.

C. Mission layer

Four UAVs execute a coverage-and-detection mission coordinated by a GCS (Fig. 2); telemetry, operator commands, and mission events all traverse the degraded interface. Coverage is credited at the GCS only from *delivered* telemetry, making mission metrics causally sensitive to transport quality. Metrics: time to completion (coverage ≥ 0.95 and all 5 victims confirmed; censored at 170 s), final coverage, completion rate.

D. Network conditions

Six conditions span the SAR envelope: *ideal*; pure loss 5/15/30%; latency 200 ms $\pm$ 50 ms jitter; and the combined condition 200 ms + 15% loss that approximates a relayed disaster link. *netem* loss is i.i.d. Bernoulli; implications are discussed in Sec. V.

E. Hypotheses

H1: under light degradation, RMW differences are operationally negligible. **H2**: under heavy pure loss the two RMWs exhibit distinct failure modes (uniform retransmission delay vs. sample shedding). **H3**: the combined condition separates the RTT distributions qualitatively (median/tail crossover). **H4**: transport degradation compresses at mission level up to an RMW-specific failure threshold.

IV. RESULTS

Table I summarizes the campaign ($2 \times 6 \times 5$ runs, 4096 B class shown).

a) Three regimes (H1 supported): Under 5% loss or pure latency+jitter the middlewares differ by tens of milliseconds and all missions complete. Under 15% loss Zenoh keeps p95 at 690 ms vs. 1056 ms at a small reliability cost (2.3% vs. 0.8%).

TABLE I
TRANSPORT AND MISSION RESULTS UNDER PHYSICAL DEGRADATION, $N=5$. LOSS = END-TO-END SAMPLE LOSS WITHIN DEADLINE; MISSIONS CENSORED AT 170 S.

Condition	CycloneDDS				rmw_zenoh			
	p95 [ms]	loss [%]	t_{mis} [s]	c.	p95 [ms]	loss [%]	t_{mis} [s]	c.
ideal	1	0.0	117	5/5	2	0.0	129	5/5
loss 5%	146	0.0	122	5/5	172	4.4	123	5/5
loss 15%	1056	0.8	135	5/5	690	2.3	128	5/5
loss 30%	2320	38.7	132	5/5	3645	35.4	164	1/5
200 ms $\pm$ 50	490	1.8	128	5/5	477	0.8	123	5/5
200 ms + 15%	2523	34.1	132	5/5	4125	21.8	133	4/5

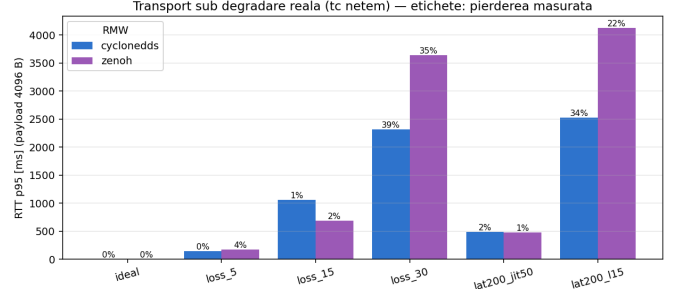


Fig. 3. Transport p95 (4096 B) with measured loss labels: light degradation is benign; heavy regimes separate the philosophies.

b) Two failure modes (H2, H3 supported): At 30% loss CycloneDDS trades delay for delivery—a uniform ~ 2.3 s wall, 5/5 missions— while Zenoh trades delivery for freshness: faster medians, 35% samples lost, 1/5 missions, coverage stalling near 0.9. In the combined condition the CDFs cross (Fig. 4): Zenoh leads the median, trails the tail (p95 4125 vs. 2523 ms), loses fewer samples overall (21.8% vs. 34.1%), and completes 4/5 vs. 5/5.

c) Compression (H4 supported): Transport p95 degrades $\times 1885$ (DDS) and $\times 2547$ (Zenoh) relative to ideal, while mission time varies $\leq 12\%$ until the failure threshold. A spatial log-distance channel with terrain profiles reproduces the effect: all 8 missions complete, T90 differs by $\leq 14\%$ between RMWs ($N=2$; differences within run-to-run variability).

V. DISCUSSION

Middleware selection for SAR is regime- and stream-dependent: for teleoperation and time-critical streams Zenoh's freshness wins until loss becomes mission-threatening; for guaranteed mission progress under heavy loss DDS's recovery wins at the cost of multi-second uniform delay. *Threats to validity*: *netem* i.i.d. loss is gentler on retransmission-based recovery than bursty radio fading (conservative for the Zenoh comparison); both endpoints share one host (no NIC/airtime effects); the ~ 11 s Zenoh startup offset (present in *ideal*) is attributable to router session establishment and excluded from per-sample statistics; mission-layer spatial results use $N=2$.

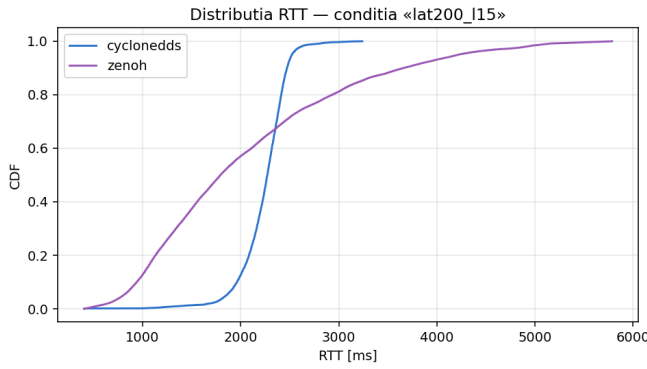


Fig. 4. RTT distributions in the combined condition: the majority of `rmw_zenoh` samples beat the CycloneDDS retransmission wall at ~ 2.3 s, at the cost of a heavier tail.

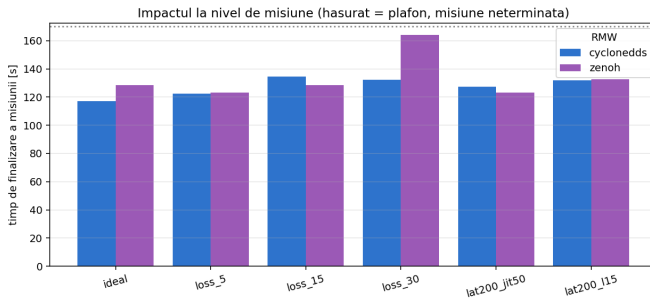


Fig. 5. Mission completion time per condition: variation $\leq 12\%$ up to the failure threshold; hatched bars denote censored (incomplete) missions.

VI. LIMITATIONS AND FUTURE WORK

Physical-testbed replication over real Wi-Fi/telemetry radios (two hosts, calibrated log-distance profiles), burst-loss models, adaptive QoS switching on link state, and a hardware tele-impedance bench extending the methodology to physical human-robot interaction.

VII. CONCLUSION

A two-layer benchmark under physical degradation reveals what transport-only studies cannot: CycloneDDS and `rmw_zenoh` implement two reliability philosophies whose failure modes differ qualitatively, and swarm autonomy compresses transport degradation by three orders of magnitude up to RMW-specific failure thresholds. Benchmark, scenarios, and data are released for reproduction.

ACKNOWLEDGMENT

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